

Candidate Profile Transformer – Technical Design

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Problem Statement

Eightfold receives candidate information from multiple heterogeneous sources such as ATS systems, recruiter exports, resumes, and recruiter notes. These sources may contain conflicting, incomplete, or malformed information. The goal is to transform these inputs into a single canonical candidate profile with normalized formats, provenance tracking, confidence scores, and configurable output schemas.

Proposed Pipeline

Input Sources (ATS JSON, Recruiter CSV, Recruiter Notes)

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Source Extraction

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Field Normalization (emails, phones, skills)

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Merge & Conflict Resolution

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Confidence Assignment

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Provenance Tracking

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Projection Layer (Runtime Config)

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Schema Validation

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Final JSON Output

Canonical Candidate Schema

candidate_id, full_name, emails, phones, location, links, headline, years_experience, skills, experience, education, provenance, overall_confidence

Normalization Rules

Email → Lowercase + trim spaces

Phone → E.164 format

Skills → Canonical skill dictionary

Dates → YYYY-MM

Country → ISO-3166 Alpha-2

Name → Trim whitespace

Merge and Conflict Resolution

Matching uses Email, Phone, and Name similarity. Source confidence: ATS=0.95, CSV=0.90, Resume=0.85, Notes=0.70. Highest-confidence source wins.

Confidence Assignment

Confidence is derived from Source Reliability, Normalization Success, and Field Completeness. Overall profile confidence is the average of merged field confidences.

Provenance Tracking

Example: {"field":"full_name","source":"ats","method":"highest_confidence"}

Runtime Configurable Output

Projection layer allows field selection, renaming, normalization, confidence/provenance toggles, and missing-value handling without code changes.

Validation Strategy

Validate required fields, types, and missing-value policies (null, omit, error).

Edge Cases Handled

Missing source, malformed source, conflicting values, invalid phone numbers, empty candidate records.

Deliberately Out of Scope

LinkedIn scraping, GitHub scraping, OCR-based resume parsing, cross-candidate identity resolution, real-time API enrichment.

Design Goals Achieved

Deterministic behavior, explainable outputs, provenance tracking, confidence scoring, schema validation, runtime configurable outputs, graceful degradation, extensible architecture.